PANJABUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Food science and Technology)

as per NEP 2020

w.e.f. session 2025-26

Course Outline for B.Voc Food science and Technology
as per NEP 2020 w.e.f. session 2025-26

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 101	Core Discipline	Introduction To Food Microbiology	3	1	4	40	10	40	10	100
FST 102	Core Discipline	Bakery And Confectionery Technology-I	3	1	4	40	10	40	10	100
FST 103	Core Discipline	Bakery And Confectionery Technology-II	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC I	Skill Enhancement Course 1	Common Basket	2	1	3	40	10	20	05	75
IDC I	Interdisciplinary Course MDSC 1	Common Basket			3	60	15			75
VAC I	Value-Added Course1	Common Basket			2	40	10			50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 201	Core Discipline	Dairy Technology	3	1	4	40	10	40	10	100
FST 202	Core Discipline	Milk And Dairy Food Products	3	1	4	40	10	40	10	100
FST 203	Core Discipline	Management of Food Industry	4		4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC II	Skill Enhancement Course 2	Common Basket	2	1	3	40	10	20	05	75
IDC II	Interdisciplinary Course MDSC2	Common Basket MDSC2			3	60	15			75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Baking Technician, Dairy Products Processor

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in Food Science and Technology = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 301	Core Discipline	Food Microbiology	3	1	4	40	10	40	10	100
FST 302	Core Discipline	Food Analysis and Instrumentation	3	1	4	40	10	40	10	100
FST 303	Core Discipline	Microbiological Analysis	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC I	Skill Enhancement Course 3	Common Basket	2	1	3	40	10	20	05	75
IDC III	Interdisciplinary Course MDSC 3	Common Basket			3	60	15			75
VAC I	Value-Added Course2	Common Basket			2	40	10			50
PW II	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 401	Core Discipline	Food Quality Testing and Evaluation	4		4	40	10	40	10	100
FST 402	Core Discipline	Technology of cereals, pulses and oilseeds	3	1	4	40	10	40	10	100
FST 403	Core Discipline	Food Safety	3	1	4	40	10	40	10	100
FST 404	Core Discipline	Food Quality Management	3	1	4					
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
VAC III	Value-Added Course3	Common Basket	2		2	40	10	20	05	75
SIT II*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Food Microbiologist

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within oneweek of start of 5th Semester. Viva-Voce examination to be held within 3 weeks of the start of 5 th semester.

ESSENTIAL COMPONENTS OF COURSE OF
B.Voc Food science and Technology
(SEMESTERS I & IV)

PROGRAMME OUTCOMES

PO1. They will be able to demonstrate knowledge of Food science, Food Microbiology, and Food Technology.

PO3. They will demonstrate an ability to identify, formulate and solve processing problems related to food industry

PO4. They will able to focus on the importance of safe processed nutritious food.

PO5. They will get exposure to various technologies aspect of Food product preparation.

PO6. They will demonstrate an ability to design or process food products as per the needs and specifications.

PO7. They will demonstrate an ability to work in Food industries, research organization and teaching.

Semester I

1	**Course Title**	**Introduction To Food Microbiology**
2	**CourseCode**	**FST 101**
3	**Typology of Course**	**Core Discipline :**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Baking Technician**

8	**Course Objectives**	**To understand the types of food microbes, causes for food spoilage, process for food spoilage, need for food preservation.**
9	**Course Outcomes**	**On successful completion of the course, the Students will get a basic knowledge about significance of microbes in different types of food, microbial growth characteristics, various methods of food preservation, hurdle technology etc.**
10	**Pedagogy of the Course Work**	Class Room, Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

Introduction to Food Microbiology

• Definition and Scope of food microbiology
• Characteristics of Microorganisms in Food
• Types of microorganisms, Classification and Nomenclature, Morphology and Structure and their importance in food (bacteria, fungi, viruses and prions, protozoan and others)
• Significance of spores
• Heat resistance of microorganisms and their spores –bacteria and their spores – fungi and their spores – yeast and their spores - spoilage of canned foods and types of spoiled cans

Unit II

• Microbial Growth in Food
• Microbial Growth Characteristics- Bacterial growth curve, microbial reproduction and microbial growth in food
• Factors affecting the growth of microorganisms in food

Unit III

Microbial Food Spoilage

• Sources of Microorganisms in foods
• Some important food spoilage bacteria
• Food spoilage: Microbial, physical, chemical & miscellaneous.
• Changes caused by micro-organisms during spoilage (breakdown of proteins, carbohydrates, fats and other constituents)
• Detection of food pathogens: Overview of Conventional and Rapid methods to detect food pathogens.

Unit IV

Control of Microorganisms in Foods

• Principles and methods of preservation
• Physical Methods of Food Preservation- Dehydration, Freezing, Cool Storage, Heat Treatment (esp. thermo bacteriology), Irradiation
• Chemical Preservatives
• Bio preservatives
• New Non Thermal methods
• Introduction to Hurdle concept and Predictive Microbiology

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. Introduction to the Basic Microbiology Laboratory Practices and Equipments
2. Functioning and use of compound microscope
3. Cleaning and sterilization of glassware
4. Preparation and sterilization of nutrient broth
5. Cultivation and sub-culturing of microbes
6. Preparation of slant, stab and plates using nutrient agar
7. Morphological study of bacteria and fungi using permanent slides
8. Methods of sterilization and preparation of media
9. Simple staining
10. Endospore staining
11. Standard Plate Count Method

13 Suggested Resources for Reading

1) Frazier William C and Westhoff, Dennis C. Food Microbiology, TMH, New Delhi, 2004
2) Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000
3) Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997
4) Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 199
5) Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.
6) Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Eastern Economy Edition, 6th edition,
7) William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, The Mc Graw-Hill Companies.

Note: Latest edition of text books may be used.

1	**Course Title**	**Bakery And Confectionery Technology-I**
2	**CourseCode**	**FST102**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Baking Technician**
8	**Course Objectives**	• To provide know about the machinery and process involved in the baking process • To understand the various types of sugar and its grades • To know the confectionery product manufacture
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. Demonstrate a comprehensive understanding of the machinery, processes, and techniques involved in bakery and confectionery production, encompassing ingredient selection, mixing methods, baking temperatures, and quality control measures. CO2. Develop technical skills in operating bakery and confectionery equipment, conducting quality tests on raw materials and finished products, and implementing hygiene and sanitation practices to ensure food safety and product quality
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

- Introduction to food processing and preservation.
- Various products of the bread and bakery sub-sector
- List the various types of industries within the bakery sub-sector
- Explain the baking process
- Equipments used in Bakery (Dough mixers, Dividers, rounders, Proofing, molding, Ovens, Slicers etc), Cleaning and maintenance of the work area and machineries, Baking ingredients required for production and plan production sequence
- Testing of Flour for Bakery Goods: Laboratory testing of Wheat grain Quality, Moisture tests, Grain hardness testing. Testing, Visco graph, Amylograph, Ferinograph. Units of measurement used in the baking industry.
- Raw material required for bakery products. Role of flour, water, yeast, salt, Sugar milk and fats.
- Yeast- An elementary knowledge of Baker's yeast, the part it plays in th fermentation of dough and conditions influencing it's working.
- Effect of over and under fermentation and under proofing of dough and other fermented goods.

Unit II

- Process of mixing and knead ingredients to make dough
- Oven & Baking – Knowledge and working of various types of oven.
- Biscuit-Types of biscuit dough – Developed dough, short dough, semi-sweet, enzyme modified dough and batters- importance of the consistency of the dough, Factors affecting the quality of biscuits/ cookies
- Cake – Flour specification – ingredients: Cake making ingredients- flour, sugar, shortening and egg types of chemically aerated goods Moistening agents Fats and Oil leavening agents
- Manufacturing process – Cake making methods sugar batter process, flour batter process, genoise method and blending and rubbing method, Correct temperature for baking different varieties of cakes, Balancing of cake formula Characteristics of cakes: external: Internal ,Types of icing.

Unit III

- Types of Confectionery, raw materials and processing of chocolates, hard boiled candies.
- Additives for Confectioneries.
- Equipments used in Confectionery manufacture
- Toffees – Milk toffee; chocolate; stick jaws; liquor chocolate.
- Baking temperatures for confectionery goods.
- Process of proofing process of baking products in the oven
- Process of cooling of baked products.

Unit IV

Bread manufacturing process – Straight dough fermentation Bread improvers- improving physical quality Methods of bread making

- Straight dough methods
- Delayed salt method
- No time dough method
- Sponge and dough method Characteristics of good bread
- External Characteristics – volume, symmetry of shape
- Internal characteristics- color, texture, aroma, clarity and elasticity Bread faults and their remedies.
- Bread diseases- Rope and Mould – causes and prevention.

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of cookies and biscuits**
2. **Preparation of different types of cakes**
3. **Preparation of different types of pastries**
4. **Preparation of different types of ice creams (Vanilla, Strawberry, chocolate, Pineapple, Mango)**
5. **Preparation different types of Pudding**
6. **Preparation of Bread rolls; Bread sticks & softs rolls;**
7. **Preparation of Fermented doughnuts;**
8. **10. Preparation of Pizza base**

13 Suggested Resources for Reading

1. Samuel A. Matz, "Bakery Technology and Engineering", Chapman & Hall, 3rd Edition, 1992.

2. Reference Books 1. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

3. Pomeranz Y, "Modern Cereal science and Technology" MVCH Publications, NY, 1987.

4. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

5. Samuel A. Matz, "Equipment for Bakers" Pan Tech International Publication, 1988.

6. Stanley P Cauvain, Linda S Young, "Technology of Bread making", Aspen publication, 2ndEdition, 2007.

7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

8. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used.

1	**Course Title**	**Bakery And Confectionery Technology-II**
2	**Course Code**	**FST 103**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Baking Technician**
8	**Course Objectives**	To understand the various types of sugar and its grades To know the confectionery product manufacture
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. Acquire problem-solving abilities to tackle challenges related to ingredient sourcing, production planning, equipment maintenance, and quality assurance in bakery and confectionery operations. CO2. Demonstrating professional responsibility, they will adhere to industry standards, regulations, and best practices, including compliance with food safety laws, hygiene standards, and waste management protocols. CO3.To provide students' knowledge about the machinery and process involved in the baking process
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

- Bakery layout- The required approvals for setting up of a Bakery- Government procedure and Bye laws.
- Selection of site & Selection of equipment
- Layout design
- Electricity
- Process of cleaning the work area and machineries after production
- Roles and responsibilities of a bakery production specialist.
- Plan, organize and management of work in bakery.
- Method of documenting and recording the details of baking ingredients to final finished product

Unit II

- Quality control of raw material and of finished products
- Role of Hygiene in Bakery
- Personal hygiene, care of skin, hand and feet, Food handlers hygiene, protective clothing.
- Dishwashing methods- manual and machine dish washing- merits and demerits
- Garbage disposal- different methods- advantages and disadvantages
- Food poisoning- Causative factors and the precautions to be taken by food handlers.

Unit III

- Food Storage- Techniques of correct storage, storage temperature of different commodities prevent bacterial manifestation or contamination. Storage of confectionery products.
- Pest Control- Rodents and insect control techniques, special stress on control of flies , rats and cockroaches, care of the premises and equipment
- Municipal health laws.
- Golden rules of first aid and treatment for cuts, wounds, burns
- Process of packaging of baked products

Unit IV

- Accounts and costing:
- Purchase book, purchase return book, Stores requisition.
- Sales book, Sales return book, Cash voucher. Credit voucher book
- Preparation of invoices and debit/ credit memos
- Percentage and discounts
- Introduction to ingredient costs, labour costs, overheads, gross profit calculation of cost price, sales price and catalogue price.
- Commodities: Sugar, Cocoa, Chocolate, Milk, Butter, Cream, Cheese, Food colours, Flavors & essences
- Dry fruits & nuts used in confectionery, Fresh and preserved fruit products
- Food laws- Agmark

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

- **Prepare the Bread rolls/cakes/ and assessment of its quality**
- **Perform the quality test on fat/butter for bakery applications on different parameters**
- **Carry out the assessment of market Candy and Chewing gum.**
- **Perform the quality assessment test on yeast and skimmed milk powder for bakery application.**
- **Prepare Chocolate cookies and assessment of its quality.**

13 Suggested Resources for Reading

1. Samuel A. Matz, "Bakery Technology and Engineering", Chapman & Hall, 3rd Edition, 1992.

2. Reference Books 1. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

3. Pomeranz Y, "Modern Cereal science and Technology" MVCH Publications, NY, 1987.

4. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

5. Samuel A. Matz, "Equipment for Bakers" Pan Tech International Publication, 1988.

6. Stanley P Cauvain, Linda S Young, "Technology of Bread making", Aspen publication, 2ndEdition, 2007.

7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

8. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used.

Semester 2

1	**Course Title**	**DAIRY TECHNOLOGY**
2	**CourseCode**	**201**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Dairy Products Processor
8	**Course Objectives**	**To know the compositional and technological aspects of milk** **To study processed milk products**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1. To understand the technology and constituents of dairy products and their preparation** **CO2. To develop technical proficiency in various areas such as milk composition analysis, dairy processing techniques, quality control measures, and sanitation protocols in dairy industry settings. Additionally, students will acquire problem-solving skills to identify, analyze, and address challenges related to milk processing, product development, quality assurance, and food safety in dairy operations.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	**Credits: 3** **Theory: 40** **Internal Assessment: 10**

Total Marks: 50
Time: 3 hours

Unit I

TECHNOLOGY OF MILK AND MILK PRODUCTS

Composition and Physical properties of milk , milk color, taste, pH and buffering capacity, refractive index, viscosity, surface tension, freezing, boiling point, specific heat, OR, electrical conductivity.

Unit II

Milk fat

• Composition and structure, factors affecting melting point, boiling point, solubility and Refractive Index, fat constants (saponification value, iodine value, RM value, Polenske value, peroxide value).
• Chemical reactions of fat (hydrolysis, auto-oxidation), condition favoring auto oxidation, prevention, measurement of auto-oxidation.

Unit III

Fats and Oils

• Classification of lipids,
• Rancidity - hydrolytic and oxidative rancidity and its prevention.
• Define - margarine, butter, hydrogenated vegetable oil, lard.

Protein and Enzymes

• General structure, amphoteric nature, difference between casein and serum protein, different types of casein (acid and rennet), uses of casein.
• Enzymes- catalase, alkaline phosphatase, lipases and proteases.

UNIT IV

• Liquid milk handling operations: Systems of collection of milk, Reception, Platform testing, Process of testing milk for accepted quality standards, organoleptic test of milk, Procedure for COB test of milk
• Various stages of processing: Filtration and Clarification, Skimming, Standardization
Homogenization, Pasteurization (LTHT, HTST, UHT) Sterilization, Packaging, Storage and distribution of fluid milk
• Description and working of clarifier, cream separator, homogenizer and plate heat exchanger. Need for processing milk.
• List the various units within a dairy processing plant
• List the machineries used in a dairy processing plant
• Roles and responsibilities of a dairy products processor in a dairy processing plant

PRACTICAL

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1. To perform platform tests in milk.(Acidity, COB,MBRT,specific gravity,SNF)
2. To estimate milk protein by Folin method.
3. To estimate milk fat by Gerber method.
4. Preparation of flavoured milk/. Pasteurization of milk
5. To prepare casein and calculate its yield.

13 Suggested Resources for Reading

1. De Sukumar, Outlines of Dairy Technology, Oxford University Press, Oxford. 2007
2. Hall GM, Fish Processing Technology, VCH Publishers Inc., NY, 1992
3. Sen DP, Advances in Fish Processing Technology, Allied Publishers Pvt. Limited 2005
4. Shahidi F and Botta JR, Seafoods: Chemistry, Processing, Technology and Quality, Blackie Academic & Professional,London,1994
5. Webb and Johnson, Fundamentals of Dairy Chemistry

Note: Latest edition of text books may be used.

1	**Course Title**	**MILK AND DAIRY FOOD PRODUCTS DEVELOPMENT**
2	**Course Code**	**202**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Dairy Products Processor
8	**Course Objectives**	**To enable the students to understand the various dairy products and their preparation.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. To apply theoretical knowledge and practical skills acquired during the course to optimize milk handling operations, improve product quality, and enhance efficiency in dairy processing plants, explore new concepts, technologies, and processes in dairy industry advancements, ethical conduct in dairy industry practices, including adherence to hygiene, sanitation, safety guidelines, and responsible waste management. CO2. To make the students aware about the diverse milk products and practically capable to develop new products for the market CO3. To demonstrate a comprehensive understanding of the compositional and technological aspects of milk, dairy products, and their development processes.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

Unit I

• Dairy industry in India & its future prospects.
• Process of production planning
• State the factors affecting operation efficiency during production
• New Concept in milk processing- UHT, Membrane processing, Microwave and irradiation treatments; aseptic packaging. Microbiology of milk, sources of milk contamination.

Unit II

• Process for producing dairy products and process of pasteurization :
• Ingredients required for production
• State the production process of pasteurization
• Process of separation and bactofugation
• Method of standardization of milk
• Method of homogenization of milk
• Method of heat exchange during pasteurization
• Process of HTST pasteurization
• Process of producing lassi, flavoured drinks, cheese dahi, kalakand, ice-cream, cream, butter , cooking butter, ghee

Unit III

• Technology of condensed and evaporated milk: Composition, nutritive value, process of manufacture, defects (their causes and prevention).
• Technology of yoghurt, Acidophilus milk, Bufgaricus milk, Kumiss, Kefir; Manufacturing of Cheddar, Mozzarella, cottage and processed cheese.
• Milk adulteration and quality control in dairy industry. By products of dairy industry and their utilization.
• Milk based infant foods; Manufacturing of casein, Caseinate, Co-precipitates, WPC, Lactose; National and International Organizations in dairy Industry.
• Milk and milk product standards and legislations in India: Grading of milk and criterion of grading, reconstituted milk, synthetic milk.
• List the different packaging materials used to pack dairy products
• Method of packaging dairy products

Unit IV

• Methods of cleaning and sanitization: Cleaning of production area, equipment, and tools used Equipment, detergents and sanitizers used in the cleaning and maintenance of the work area, Properties of the cleaning agents used, CIP method of cleaning,
• Lubrication system and its process followed in the dairy industry
• State the different types of maintenance procedures, Periodic maintenance of all production machineries
• Method of managing and disposing waste material
• Personal hygiene and sanitation guidelines
• Demonstrate how to use tools safely
• Food safety hygiene and quality standards to follow in a work environment , HACCP principles to eliminate food safety hazards in the process and products
• Method of documenting and recording the details of raw material to final finished product

PRACTICAL

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1 Platform tests,
2 Determination of fat, SNF, TS Protein,
3 Lactose and ash contents of milk.
4 Layoput plan for setting up of milk plant,
5 Preparation and Evaluation of different types of milk and milk products 6 Preparation and evaluation of butter, ice cream, cheese, yoghurt

13 Suggested Resources for Reading

1. **Sukumar. 1983. Outlines of Dairy Technology. De,Oxford University Press.**
2. **James N.Warner.1989. Principles of Dairy Processing by, Wiley Eastern Ltd.**
3. **Eckles,Combs; and Macy. 1982. Milk and Milk Products by,Tata McGraw Hill.**
4. **Aneja et al. 2002.Technology of Indian Milk Products.A Dairy India Publication.**

Note: Latest edition of text books may be used.

1	**Course Title**	**MANAGEMENT OF FOOD INDUSTRY**
2	**Course Code**	**FST 203**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory +00 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Dairy Products Processor**
8	**Course Objectives**	**To develop a knowledge base in key areas of institutional food management.** **To impart necessary expertise to run a food service unit.** **To provide practical level experience in managing food service management.** **To critically evaluate the functioning of food service units.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. To impart necessary expertise to run a food service unit. CO2. To critically evaluates the functioning of food service units. CO3 To provide practical level experience in managing food service management
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100
Time: 3 hours

Unit-I

• Institutional management, Levels of Management.
• Review of different types of food service operations. Commercial, non- commercial food service institutions.
• Importance of food standards. Need for knowledge as to what constitutes good products, criteria for good quality products, hazards of poor quality, quality control, Specification for quality in food products.
• Organization - definition and types

Unit-II

•The Manufacturing (Production) Function: Objectives of Production Management, Operation Concept, Meaning of Product Development, Factors Influencing Choice of Manufacturing Systems, Classification of Manufacturing Systems;
• Basic Procedure of Method Study (Work Study) and Time Study; Concepts, Objectives and functions of Production Planning and Control (PPC).
• Physical plant- its location, floor plans, space allowance, Institutional food units-storage, preparation and serving
Cost control
a) Food cost, Labour cost, Maintenance cost
b) Budgets
c) Documentation and Records: Documentation and maintenance of record of raw materials and packaging materials
d) Portion control
• Role of supervisor in fruits and vegetable processing industry. • Planning and organization of work: organization standards, process standards and procedures followed in the organization, types of products produced by the organization, Code of business conduct, Dress code.

Unit-III

• Personnel Management
• Personnel and leadership qualities for food service administration.
• Labour , Types of labour, criteria for selection and employees training.
• Labour laws and legal aspects- health & safety of employees, welfare policies.
• Manage and lead a team Management Skills-The Importance of Delegation, Developing Your Team, Motivating Your Team: Motivation Theories – Maslow's, Herzberg's; Worker's Participation in Management; Collective Bargaining.
• Communicating and Working With Your Team – and With Others, Communicating With People Outside Your Team
• Managing Discipline

Unit-IV

Hygiene and Sanitation
i) Personal hygiene- Importance of personal cleanliness in handling and serving food. Health examination of personnel. ii) Sanitation in handling food and equipment.
a. Preventing contamination of cooked food and handling of fresh foods,. b. Cleaning and hygienic handling of equipments. c. Insect and Rodent control d. Safety, general safety rules in food preparation and service area. Accident prevention
Equipment- Types of equipment- criteria for selection, operation and care.
Food laws and regulations on raw materials, product(s), packaging and labelling
Food safety, hygiene, sanitation GMP , HACCP and quality assurance procedures
Procedures for disposal of waste.

13 Suggested Resources for Reading

1 West Wood A; Harper Food Service in Institution.
2. West, Bessin, Brooks; Food Service in Institutions
3. A.M. Home Economics Association; Hand Book of Food Preparations:
4. Sweetman, M.M. 4, Mac, Keller; Food Selection and Preparation:
5. Oliver B., Watson;School Lunch Room Service
6. Lender H. Katshever and Margret E. Terrel; Food Service Planning: Layout Equipment
7. Davidson and Passmore- Human Nutrition and Dietetics.

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Microbiology**
2	**Course Code**	**301**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) e. The exam of this course will be of 3hours duration. f. There will be no objective type questions. g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	**To introduce to students fundamental concepts of microbiology**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO. **To understand the fundamental concepts of microbiology that includes microbial diversity, their morphological and functional properties** **CO2: To understand the techniques to study microorganisms, growth and control of microorganisms as well as food spoilage and its control.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

UNIT I

- **Introduction – Discovery of microbial world, Importance of microbiology, Microbial classification and nomenclature, Characteristics of major groups of microorganisms: Archaebacteria, Eubacteria, Fungi, Protozoa and Viruses and Bacteriophages.**
- **Principles of food microbiology and microbiology terminologies**
- **Significance of micro-organisms in preservation, production and spoilage of food**
- **Microbial Nutrition: Nutritional requirements of microbes; Types of culture media; Classification of microbes on the basis of nutritional requirements, Identification of bacteria.**

UNIT II

- **Human-Microbial Interactions: Normal flora –Gastrointestinal tract; Pathogenic mechanisms of food borne bacteria, Brief account of mechanisms of action of chemotherapeutic agents, Introduction to specific and nonspecific defense mechanisms to infections.**
- **Enumeration of Microorganisms- qualitative and quantitative**
- **Food Fermentations**
- **Fermentation –definition and types**
- **Microorganisms used in food fermentations**
- **Dairy Fermentations-starter cultures, types and methods of preservation and propagation, Lactic acid and aroma compounds production, Health benefits of LAB, probiotics, prebiotics and symbiotic**
- **Fermentated Foods-types, methods of manufacture for vinegar, sauerkraut, tempeh, miso , soya sauce ,beer, wine and traditional indian foods**

Unit III

Food borne Diseases

- **Types – food borne infections, food borne intoxications and toxicinfections**
- **Origin, General characteristics , symptoms and prevention of some commonly occurring food borne diseases caused by Bacillus spp., by Clostridium botulinum,Staphylococcusaureus; Clostridium botulinum; C. perfringen; Listeria monocytogene; Salmonella; Escherichia.coli; Yersinia enterocolitica; Shigella spp., Vibrio parahaemolyticus, Mycotoxins, Hepatatis, Gastroenteritis viruses, Entamoebahistolytica and Entamoeba coli..**
- **Emerging pathogens of concern**

Unit IV

- **Microbiology of Water and Food Commodities: Microbiology of water and their importance of processing of foods in industries. MPN of coliforms Microbiology of milk – Hetero and homo fermentative Lactic acid bacteria – Yogurt and Cheese fermenting**

Trends in Food Microbiology

- **Rapid Methods of Detection**
- **SCP and SCO**

Recent Advances

- **Microbial quality control: Determination of microorganisms in foods by cultural, microscopic, physical, chemical methods.**

Practical

Credits: 1

Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1. **Introduction and study of microbiological instruments.**
2. **Media preparation, aseptic techniques and transfer of microorganisms.**
3. **To study various culture techniques- pour plating, spread plating and streaking.**
4. **To study morphology of bacteria by simple staining and negative staining.**
5. **Principle, procedure and use of gram staining method.**
6. **Staining of bacterial spores.**
7. **To distinguish the growth characteristics of microorganisms in various differential and selective media.**
8. **Identification of fungi by Lactophenol cotton blue staining method.**
9. **Sampling and observation of microorganisms from natural sources.**
10. **To study serial dilutions of the sample and plate counts.**
11. **Standard plate count for raw, pasteurized milk and ice-cream.**
12. **Presumptive coliform test for milk, butter and ice-cream.**

13 Suggested Resources for Reading

1. Tortora G.J et al, 2013, Microbiology: an introduction, Pearson Education
2. Nester, E.W, 2011, Microbiology, McGraw-Hill Higher Education
3. Dubey, R.C., Maheshwari, D.K., 2011, Textbook of Microbiology, S.Chand Publications, ND
4. Adams, M.R., Moss, M.O., 2012, Food Microbiology, New Age International Pvt. Ltd., ND.
5. Pelczar, Reid and Chan, 2008, Microbiology, McGraw hill Ed, ND
6. Brock biology of Microorganisms, 2011, Benjamin Cummings publishers.
7. Ingraham, John.L.2009, Introduction to Microbiology, 3 Ed., Thomson brocks/Cole Inc.
8. Ananthanarayan, Panikar, CKJ.,2013, Textbook of Microbiology, Oriental Longman Pvt. Ltd., Hyderabad.
9. Frazier, William, C. 2008, Food Microbiology, Tata McGraw Hill Ed., ND.

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD ANALYSIS AND INSTRUMENTATION**
2	**Course Code**	**302**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i. The exam of this course will be of 3hours duration. j. There will be no objective type questions. k. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	**To generate the skill of handling the different instruments food analysis.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO1: **To handle the different instruments of food process technology.** **CO2: To study the various techniques of food analysis.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

Unit 1

- **Introduction to food analysis, type of samples and sampling techniques, storage and preservation of samples, expression of results. Different preservation principles involved in food processing Sampling: Sampling techniques and preparation of food samples. Sampling of Milk, Eggs, Dried &Frozen food products.**
- **Physico chemical methods for food analysis: Moisture & Total solids Carbohydrates Proteins Fats Fiber Ash & its types Minerals Vitamins Enzymatic methods**
- **Sensory Tests: Difference, Rating & Sensitivity tests. Types of panels Testing area & schedule**

Unit II

- **Instrumentation in food analysis : Principles, types and applications of colorimetry and electrophoresis and chromatography :paper, thin layer, ion exchange**

Unit III

- **Instrumentation in food analysis: Color measurement in foods. X-ray analysis of foods and its applications, scanning electron microscopy (SEM) in food analysis and identification**

Unit IV

- **Ultrasonic and other instruments for determination of physical and rheological properties of food Texture analysis in foods.**
- **Sensory versus instrumental analysis of texture, rapid methods of microbial analysis, immunoassays; Techniques for estimation and analysis of toxins and pesticides in food.**

Practical:

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1. **Use of spectrophotometer in food content estimations.**
2. **Determination of physical and rheological properties of foods by different techniques.**
3. **Separation of food components by chromatography.**

13 Suggested Resources for Reading

1. **SemihOtles. 2008. Handbook of Food Analysis Instruments. CRC Press**
2. **Gulaboski, Rubin and Carlos M, Pereira (2008) Electrochemical techniques and instrumentation in Food Analysis in Handbook of Food Analysis Instruments (book, SemihOtles, ed.). In: Handbook of Food Analysis Instruments. Wiley.Practicals**
3. **Nielsen, S.S, 2004, Introduction to chemical Analysis of foods, CBS Publishers, New Delhi.**
4. **Ranganna. S., 2001, Handbook of Analysis & Quality control for Fruit & Vegetable Products, Tata McGraw Hill, New Delhi.**
5. **Pomeranz.Y, Meloan.C.E, 1996, Food Analysis – Theory & Practice, CBS Publiushers, New Delhi**
6. **Jacobs.M.B., 1999, Chemical Analysis of Food & Food Products, CBS Publiishers, New Delhi.**
7. **Jay.J.M, 1996, Modern Food Microbiology, CBS Publishers, New Delhi.**
8. **Debnath, 2005, Tools & Techniques of Biotechnology, Pointer Publishers, Jaipur**

Note: Latest edition of text books may be used.

1	Course Title	Microbiological Analysis
2	**Course Code**	**303**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) m. The exam of this course will be of 3hours duration. n. There will be no objective type questions. o. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. p. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To understand the skill of handling the different instruments of microbiological analysis**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO1: **To handle the different instruments of microbiological analysis** • **To understand the various techniques of microbiological analysis**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
11	**Instructions for paper setters and candidates**	**5. The syllabus of this paper has been divided into FOUR units.** **6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **8. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

Unit I

- **Equipments and tools used for of food products such as weighing scale, homogenizer, autoclave, laminar air flow chamber, vacuum pump, Bunsen burner, inoculation loop, incubator, refrigerator, microscope, etc.**
- **Microbiological techniques for analysis of food products**
- **Analyze results and provide recommendations SOPs for preparing culture media,**
- **Prepare liquid and solid culture media (nutrient broth and nutrient agar) following SOP and prepare for sterilization**

Unit II

- **Sterilization of culture media, solvents and glassware**
- **Disposal of used microbiological media and culture**
- **Quality standards for food products produced in the organization and methods to assess and maintain quality of food products**
- **GMP**

Unit III

- **Cleaning process to disinfect equipments/ tools and glass wares used ,Recommended sanitizers**
- **Organization standards to clean laminar air flow cabinet or lab bench using approved disinfectants and sanitizers, health and safety, recommended dosage for use of sanitizers, control of substances hazardous to health, handling/storage/ disposal/ cautions of use of sanitizers and disinfectants, fire precautions, occurrences, hygiene practice, disposal of waste, environmental protection, etc.**
- **Standards and procedures followed in the organization for cleaning and disinfecting equipment/ tools in microbiology laboratory**

Unit IV

- **Job responsibilities/duties to maintain lab hygiene standards.**
- **Dress code .**
- **Hygiene, safety and quality standards.**
- **Hygiene requirements and standards relevant to microbiology laboratory.**
- **Food Safety Standards and Regulations (as per FSSAI), food safety systems like HACCP, food hygiene and sanitation.**
- **Knowledge of legal regulations pertaining to work place- food regulatory system such as FSSAI and food standards for products produced in the organization.**
- **Microbiological hazards in production process and its critical control point to minimize or prevent those hazards.**
- **Physical, chemical and biological hazards and methods of prevention of various hazards**
- **Personal hygiene.**
- **Document and maintain records on microbiological analysis of raw materials for production of food products.**
- **Management and leading of team.**

Practical

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1. **Quality audit methods and procedures**
2. **lactophenol staining**
3. **Hanging drop preparation to observe motility of bacteria**

13 Suggested Resources for Reading

1. **Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.**
2. **Frasier.1987. Food Microbiology.Tata McGraw-Hill Education Reference Books 1.Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Easterrn Economy Edition, 6th edition,**
3. **William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, TheMcGraw-Hill Companies,**
4. **Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000**
5. **Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997**
6. **Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 1993**

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD QUALITY TESTING AND EVALUATION**
2	**CourseCode**	**401**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 00 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) e. The exam of this course will be of 3hours duration. f. There will be no objective type questions. g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To understand basic quality attributes of foods and their application in industry.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. **To understand basic quality attributes of foods in raw as well as processed form.** **CO2. To understand the various systems of objective and subjective evaluation and their application in industry.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**5. The syllabus of this paper has been divided into FOUR units.** **6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **8. All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

UNIT: 1:

Introduction to quality attributes: Appearance, flavor, textural factors and additional quality factors.

Taste

- **Introduction**
- **organs involved in taste perception- tongue, papillae, taste buds, salivary glands**
- **mechanism of taste perception**
- **chemicals responsible for sweet, salt, sour, and bitter taste their structure and chemical dimensions**
- **Factors affecting taste quality, reaction time and factors affecting it absolute and recognition threshold, taste abnormalities**

UNIT II: Olfaction

- **Introduction and definition, anatomy of nose, mechanism of odour perception, Prerequisites for odour perception, odour classification, chemical specificity of odour, measurement of odour using different techniques – primitive, double tube olfactometer, Elseberg techniques, Wenzel's olfactometer, sniffing, merits and demerits of each methods, olfactory abnormalities.**

UNIT III: Colour

- **Introduction to natural and synthetic colours**
- **functions of colour in foods**
- **Optical aspect of colour,**
- **perception of colour,**
- **objective evaluation, colour measurement using different systems**
- **Qualitative and quantitative analysis of colour,**
- **Reflectance spectrophotometry and Colorimetry.**

UNIT IV: Texture

- **Introduction**
- **Definition and classification of texture profile**
- **Subjective evaluation, phases of oral processing**
- **Objective analysis, rheological methods of texture measurement including rheological models**
- **Measurement of texture in various food groups viz. cereals, dairy, fruits and vegetables, meat and meat products**

13 Suggested Resources for Reading

1. **Amerine,Pangborn&Roessler, Principles of sensory evaluation of food, Academic Press, London,1965**
2. **DeMan, 3rd edition,Principles of Food Chemistry, Springer,2007.**
3. **Meilgard, Sensory evaluation Techniques, 3rd ed CRC Press LLC, 1999**
4. **YeshajahuPomeranz& Clifton E. Meloan, Food Analysis & Theory & Practice,1st Indian ed. CBS Publisher & Distributors, New Delhi, 2002**

Note: Latest edition of text books may be used.

1	**Course Title**	**TECHNOLOGY OF CEREALS, PULSES AND OILSEEDS**
2	**CourseCode**	**402**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i. The exam of this course will be of 3hours duration. j. There will be no objective type questions. k. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To create awareness about the processing of major cereals like paddy, maize etc.** • **To study the storage and handling techniques of cereals.** • **To gain knowledge on processing and milling of pulses.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to • **To understand the technology of milling of various cereals** • **To understand the techniques of pulses and oilseeds refining** • **To understand the byproducts obtained during processing along with their uses.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**9. The syllabus of this paper has been divided into FOUR units.** **10. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **11. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **12. All questions carry equal marks.**
12	**Course Content**	**Credits: 3** **Theory: 40** **Internal Assessment: 10** **Total Marks: 50** **UNIT 1: Technology of cereals** • **Wheat --Types, milling, flour grade, flour treatments (bleaching, maturing), flour for various purposes, technology of dough development. Chapter4-7, Kent** • **Paddy Processing: Composition of paddy, physicochemical properties and Quality characteristics. Curing of Paddy – Parboiling Processes. Physico-chemical changes during parboiling – Cold Water soaking and Hot water soaking processes. Methods of grain drying- LSU Dryer, By Products of Paddy Processing - Paddy husk**

and its uses as husk ash, activated carbon, furfural and other byproducts. Production of Flattened and Puffed Rice from Paddy.

- **Rice Milling: (mechanical & solvent extraction), Paddy Dehusking Processes. Rice Milling Flow Chart. Engelberg Huller Mills. Modern Rice Mills – their Components - Pre Cleaners, Shellers, Under Runner Shellers and Centrifugal Shellers. Paddy Separators – Satake and Schule Designs, Polishers -Cone polishers and other types, Bran and Brokens separators. Rice Mill yields and loss due to brokens at different stages of milling. Use of Rice Bran in Edible oil Industry.**

UNIT II:

- **Barley- Milling (pearl barley, barley flakes & flour), beer preparation**
- **Oats – Milling (oatmeal, oat flour & oat flakes)**
- **Sorghum and millets – Traditional & commercial milling (dry &wet)**

UNIT III:

- **Milling Of Pulses: Major Pulses grown in the country .Traditional milling process -merits and demerits. Dry milling, Wet milling & Improved milling method**
- **Drying of legumes - Sun and Traditional drying processes steps – Precleaning, Pitting, Oil application, conditioning, dehusking and splitting - Machinery and equipment employed. Modern milling process - Process flow chart -Mechanical hot air drying and conditioning - merits and demerits, dehusking in Pulse Pearler, Water conditioning, splitting of pulses in Pulse splitter, Merits and demerits. Mini dhal mill -working principle - advantages and disadvantages. Grinding of split pulses, pulse flour products, their applications and equipments used.**

UNIT IV:

- **Milling And Processing Of Maize: Products of milling - Flour, Semolina, Brewers' grits etc and their applications. Gluten and Starch Separation, Equipment needed for Degermination, Debraning and starch separation. Starch conversion into other value added products – Acid Hydrolysis, Enzyme Hydrolysis, Isomerization processes.Processing for Dextrose, Malto Dextrin and other products. Extraction and refining of Corn oil in brief. Grain storage and handling**

Technology of Oilseeds

- **Introduction**
- **Sources of protein (defatted flour, protein concentrates and isolates), properties and uses, protein texturization, fiber spinning**
- **Breakfast cereals – porridge , flakes and puffed products.**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Determination of physical properties of different cereal grains.**
2. **Determination of water absorption capacity, sedimentation value and alcoholic acidity of the Maida.**
3. **Determination of adulterant (NaHCO3) in wheat flour/ Maida.**
4. **Estimation of Protein content of different Cereals and Legumes.**
5. **Determination of Gluten content in wheat flour samples.**
6. **Physical characteristics of wheat.**
7. **Moisture content of wheat and products.**
8. **Yeast fermenting power.**
9. **Physical characteristics of rice.**
10. **Cooking quality of rice.**

13 Suggested Resources for Reading

1. Chakraverty, A.: Post Harvest Technology of Cereals, Pulses and Oilseeds. Oxford and IBH Publishing Co, Calcutta, 1995.

2. Samuel A .Matz: "The Chemistry and Technology of Cereals as Food and Feed", Chapman and Hall, 1992. Reference Books 1.Smartt J, "Tropical Pulses", Longman Group Ltd. London, 1980.

3. Pomeranz Y, "Modern Cereal Science and Technology" VCH Publishing Inc. New York, 1987.

4. Cryde M. Christensen, "Storage of Cereal Grains and their Products", American Association of Cereal Chemists

inc., St. Paul, USA, 1982. 5
5. Bernard Godon and Claude Willm, "Primary Processing of Cereals" Berns and Noble Publishers, 1994.6.KarelKulp and Joseph P Pante, "Handbook of Cereal Science and Technology", MercelDekkar, USA, 2000.
6.Kent, Technology of Cereal, 5th Ed. Pergamon Press, 2003
7. Marshall, Rice Science and Technology, Wadsworth Ed., Marcel Dekker, NewYork, 1994

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD SAFETY**
2	**CourseCode**	**403**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) m. The exam of this course will be of 3hours duration. n. There will be no objective type questions. o. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. p. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To understand the concerns of Food safety and hygiene**
9	**Course Outcomes**	On successful completion of the course, the Students will be able • **To understand the types of hazards associated with food** • **To understand the food regulations (national as well as international)** • **To design and implementation of food safety management systems such as ISO series, HACCP and its prerequisites such as GMP, GHP etc.** • **To understand the emerging concerns of food safety and hygiene.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**13. The syllabus of this paper has been divided into FOUR units.** **14. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **15. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **16. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I:

Introduction to Food Safety

- **Definition**
- **Factors affecting Food Safety**
- **Importance of Safe Foods**

Food Hazards of Physical and Chemical Origin

- **Introduction**
- **Physical Hazards with common examples**
- **Chemical Hazards(naturally occurring ,environmental and intentionally added)**
- **Impact on health**

Food Hazards of Biological Origin

- **Introduction**
- **Indicator Organisms**
- **Food borne pathogens: bacteria**
- **Food borne pathogens: viruses**
- **Food borne pathogens: eukaryotes**
- **Seafood and Shell fish poisoning**
- **Mycotoxins**

UNIT II:

Management of hazards

- **Need**
- **Control of parameters**
- **Temperature control**
- **Food storage**
- **Product design**

Hygiene and Sanitation in Food Service Establishments

- **Introduction**
- **Sources of contamination**
- **Control methods using physical and chemical agents**
- **Pest and Rodent Control**

Unit III

Food Safety Management Tools

- **Basic concept only**

Microbiological criteria

- **MRA**
- **Microbiological standards and limits (for processed food, water)**
- **Microbiological Assessment and categories of food based on microbial quality**
- **Sampling**
- **Basic steps in detection of food borne pathogens**
- **Water Analysis**

UNIT IV:

Food laws and Standards

- **Indian Food Regulatory Regime**
- **Global Scenario**
- **Other laws and standards related to food**

Recent concerns

- **New and Emerging Pathogens**
- **Packaging ,Product labeling and Nutritional labeling**
- **Genetically modified foods \ Transgenic**
- **Organic foods**
- **Newer approaches to food safety**
- **Recent Outbreaks**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of different types of media (complex, differential and selective)shift to microbiology**
2. **Microbiological Examination of different food samples**
3. **Bacteriological Analysis of Water**
4. **Assessment of surface sanitation by swab/rinse method**
5. **Assessment of personal hygiene**
6. **Scheme for the detection of food borne pathogens**
7. **Implementation of FSMS – HACCP, ISO : 22000**

13 Suggested Resources for Reading

1. **Lawley, R., Curtis L. and Davis,J. The Food Safety Hazard Guidebook , RSC publishing, 2004**
1. **De Vries. Food Safety and Toxicity, CRC, New York, 1997**
2. **Marriott, Norman G. Principles of Food Sanitation, AVI, New York, 1985**
3. **Forsythe, S J. Microbiology of Safe Food, Blackwell Science, Oxford, 2000 &Sons; USA, 1987**

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD QUALITY MANAGEMENT**
2	**CourseCode**	**404**
3	**Typologyof Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) q. The exam of this course will be of 3hours duration. r. There will be no objective type questions. s. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. t. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To learn about quality management in food production chain.** • **To learn about physical, chemical contaminants in foods** • **To learn about latest trends and techniques in food science**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to • **To understand the significance of safe processing of foods.** • **To understand the role of food standards and regulations in maintaining food quality.** • **To understand the different principles of food quality control.** • **To assess the food quality assurance of different food.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	17. **The syllabus of this paper has been divided into FOUR units.** 18. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 19. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 20. **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

Food Quality

- **Introduction to food quality management – Definition, quality concepts, quality, quality perception, quality attributes, safety, health, sensory, shelf life, convenience, extrinsic attributes, factors affecting food behavior. Principles of food quality control and quality assurance Objectives, importance and functions of quality control. Quality management principles and methods, tools and techniques.**
- **Organization policies and goals, quality management, budget management, food regulatory policies and procedures related to products produced in the organization, quality mark accreditations of the organizations, audit procedures, code of business conduct, leadership techniques, manpower modeling and handling**

- **Quality in the Agri- food production chain-Techno- managerial approach, food quality relationship and food quality management functions. Dynamics on the agri- food production chain, core developments in food quality management.**

UNIT II

Food contamination

- **Contamination in Food- : Physical, chemical (heavy metals, pesticide residues, antibiotics, veterinary drug residues, dioxins, environmental pollutants, radionucleotides, solvent residues, chemicals) Natural toxins.**
- **Total quality management (TQM) - good manufacturing practices, good hygenic practices, good lab practices Microbial quality control : determination of microorganisms in foods by cultural, microscopic, physical, chemical, immunological and bioassay methods**

UNIT III

- **Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of milk, tea, coffee, meat, spices, cereals, oils & fats and other products. Their analysis and analysis of common preservatives used in processed foods. Permissible limits of Preservatives.**
- **Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control**
- **Introduction, need of food additives in food processing and preservation. Characteristics and classification of food additives.**
- **Antimicrobial agents. -Nitrites, sulphides, sulphur di oxide, sodium chloride, hydrogen peroxide.**
- **Antioxidants - Introduction, mechanism of action, natural and synthetic anti-oxidants, technological aspect of antioxidants.**
- **Colors- Introduction, importance, classification- natural, artificial, and natural identical, FD&C Dyes and Lakes. Use of plant tissue culture, polymeric colors etc for color**
- **Quality control of following food products : Milk & milk products Oils & Fats Cereal grains & flours Fruits & vegetable products Canned foods Egg & egg products Meat & Meat products 8.**

Unit IV

- **Food Laws, standards and regulations: History, National and International laws & Regulations: USFDA, EU, USFDA, ISO 9000 series etc. Food laws and standards, IPR and patents, Food standards and safety Act : salient provisions and prospects, role of various national and international agencies**
- **World Trade Organization (Sanitary and Phyto Sanitary agreement, Technical Barriers in Trade), -Standards of Identity, Standards of fill of the container. Statistical quality control in food industry Food safety and Standards Act 2006: Salient provision and prospects, Role of national and international regulatory agencies, Bureau of Indian Standards (BIS), AGMARK, Food Safety and Standards Authority of India (FSSAI**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Qualitative tests for fats and oils, spices and condiments.**
1. **Estimation of residual sulphur dioxide in beverages.**
2. **Chromatographic estimation of colour.**
3. **Analysis of edible common salt for MC, MIW and total chlorides.**
4. **Estimation of pesticide residues in food/water.**
5. **Estimation of benzoic acid in foods.**
6. **Techniques of quality assessment of different natural and processed foods.**
7. **Identification and ranking of food product attributes**
8. **Sensory and instrumental methods for measuring food quality assessment of fruits, vegetable, dairy products, milk and other processed products.**

13 Suggested Resources for Reading

1. **Pieternel A, Luning, Willem J. Marcelis, Food Quality Management, Technological and Managerial principles and practices, Wageningen,2009.**
2. **Brannen and etal,Food Additives, Marcel Dekker,New York,1990**
3. **Shalton, Principles and Practices for the safe processing of Foods.**
4. **DeMan, 3rd edition, Principles of Food Chemistry, Springer, 2007.**
5. **InteazAlli (2003). Food Quality Assurance Principles and Practices. CRC Press .**
6. **J Andres Vasconcellos (2003) Quality Assurance for the Food Industry. A Practical Approach.. CRC Press**

Note: Latest edition of text books may be used.